Assignment 4. Forms and Validation

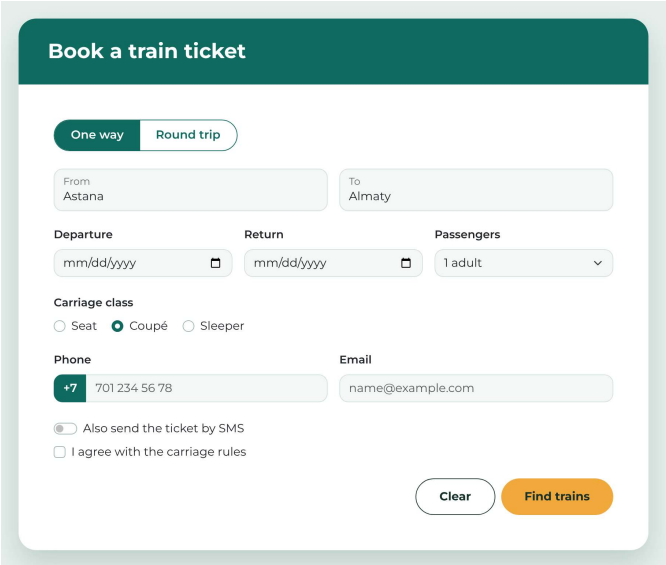
Zhumazhan Miras · IT-2512 · 18 October 2026

GitHub: https://github.com/mmionya/web-technologies-assignment4

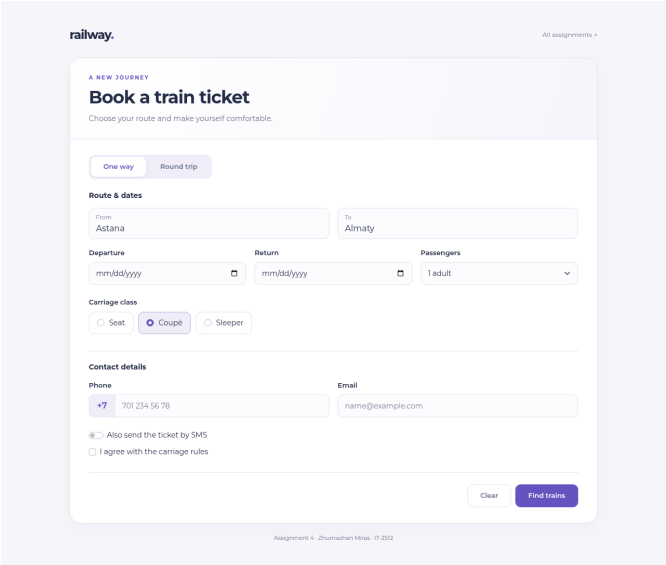
GitHub Pages: https://mmionya.github.io/web-technologies-assignment4/

Task 1. Train ticket booking - 1280 px

Assignment reference



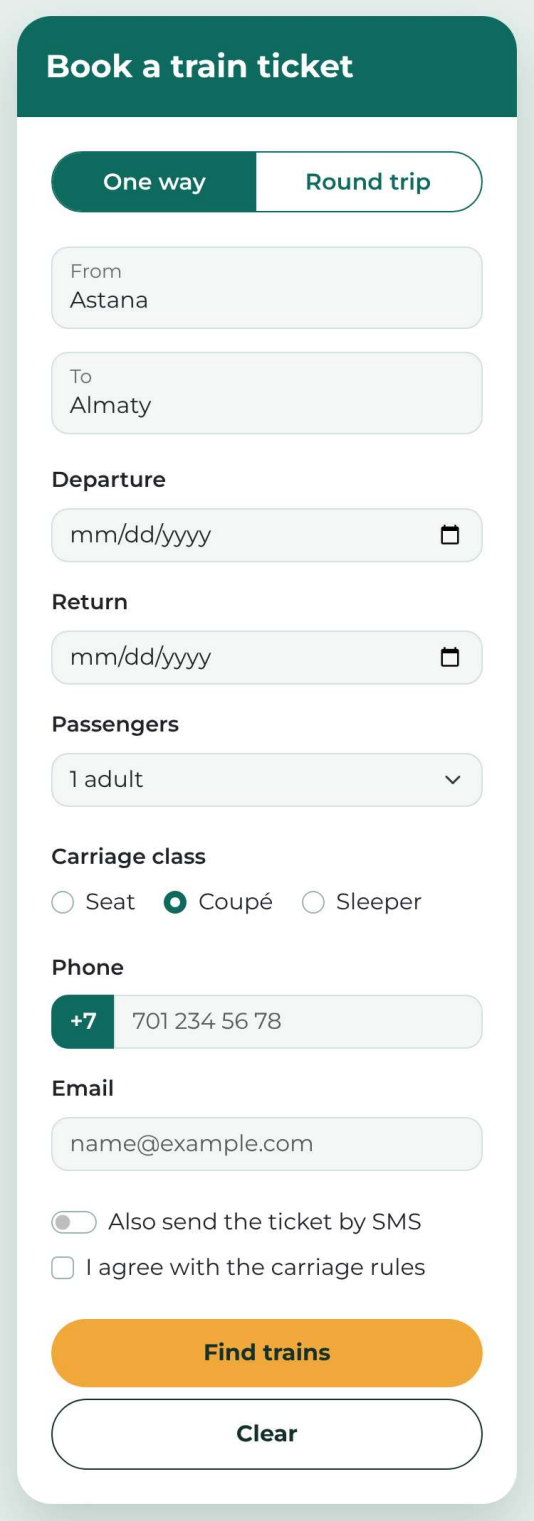
My page · 1280 px



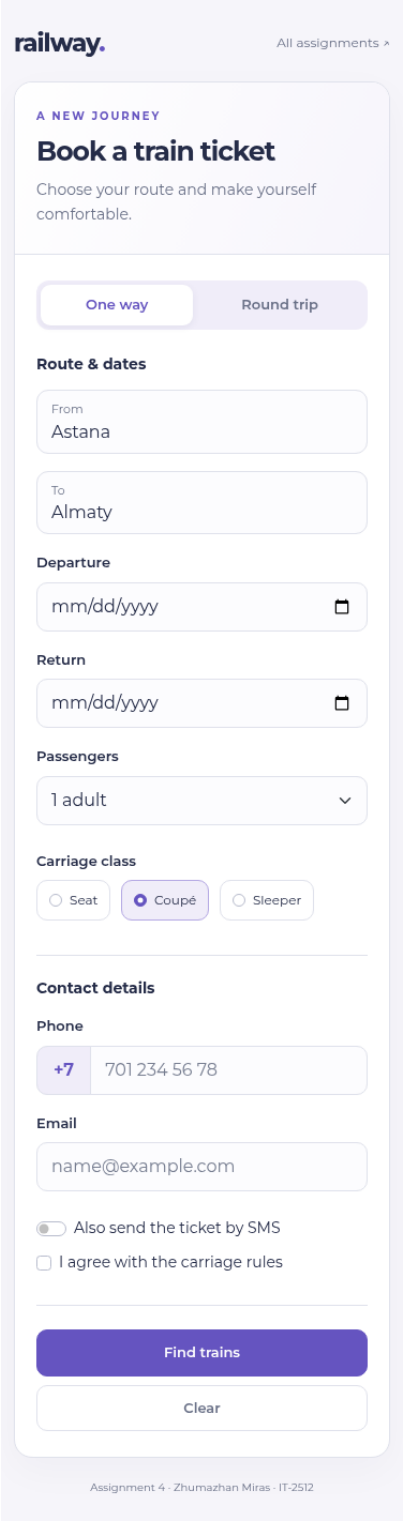
The form uses the Bootstrap grid, radio buttons, a select menu, an input group, a switch, and a checkbox. The custom navy and lavender palette uses #252C48, #6554C0, #F0EDF9, and #F6F5FA. The colors and layout differ from the reference by design; Montserrat and the Bootstrap components are retained. For comparison, One way, Coupé, and 1 adult are selected, and the route is entered manually.

Task 1. Mobile layout - 375 px

Assignment reference



My page · 375 px

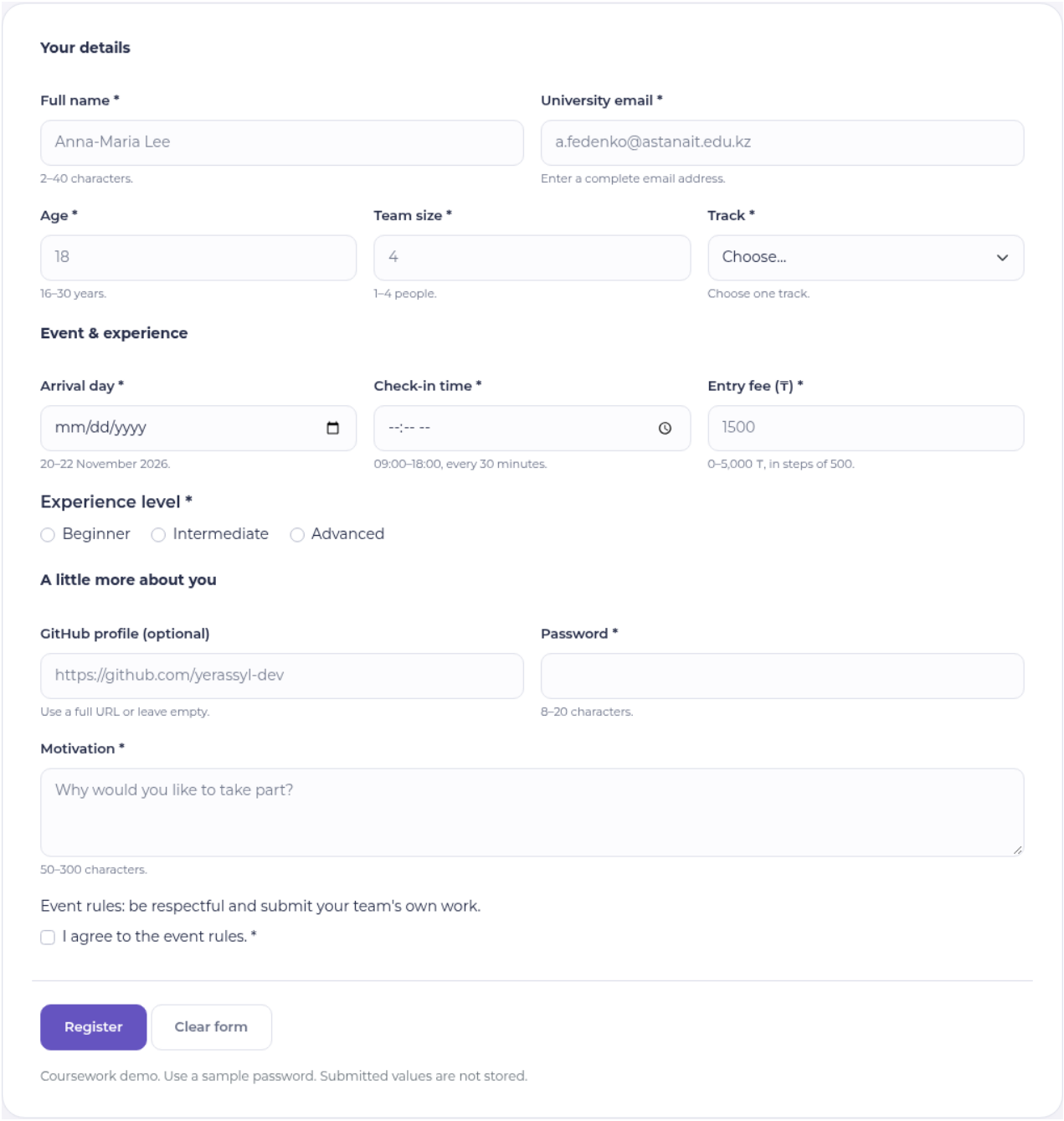


On mobile, the fields are stacked vertically. flex-column-reverse places Find trains above Clear, while flex-md-row restores the horizontal button layout on desktop.

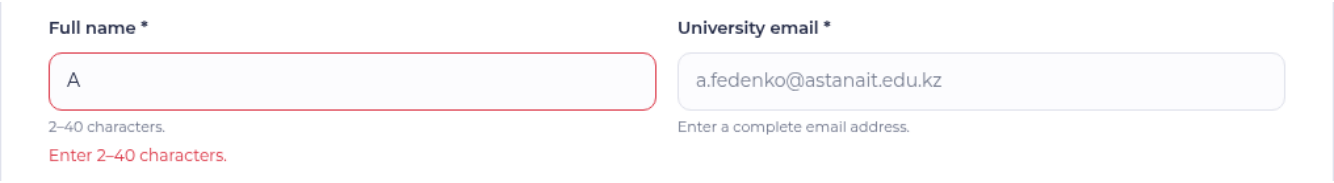
Mandatory fields use required. Passengers allows 1-6 adults. Clear uses type=reset and empties the form because its initial values are empty. Every field has an associated label; Return is optional.

Task 2. AITU Hack 2026 registration

All rules use HTML attributes. :user-invalid and .invalid-feedback display errors after interaction. No red error messages appear when the page first opens.



Example after entering a name that is too short:



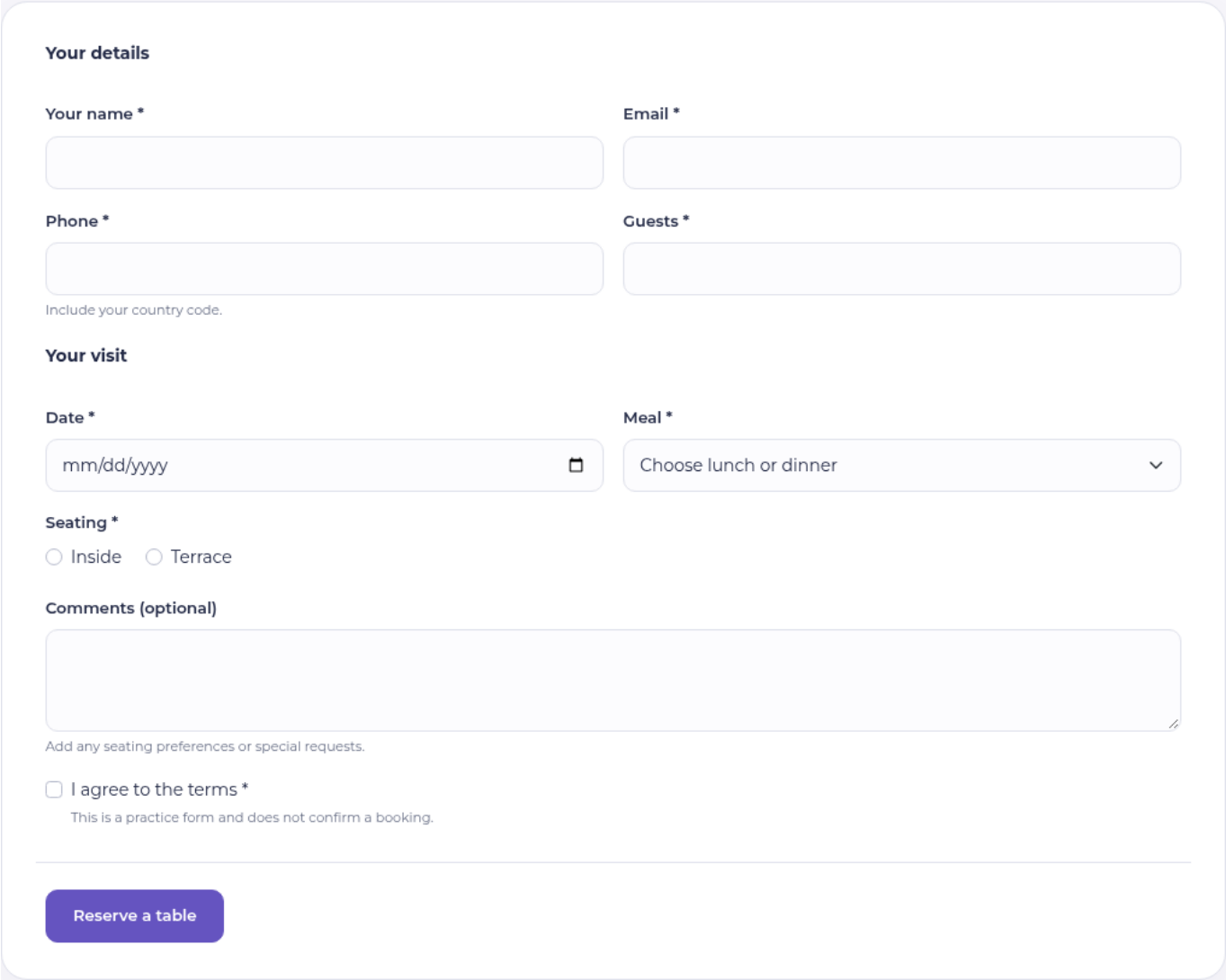
Task 2. Validation results

Field	Rule	Accepted	Rejected	Result
Full name	required; minlength=2; maxlength=40	Anna-Maria Lee	A	Passed
University email	email; required	a.fedenko@astanait.edu.kz	a.fedenko; a.fedenko@	Passed
Age	number; 16-30; step=1; required	18	15; 31	Passed
Team size	number; 1-4; step=1; required	4	5; 2.5	Passed
Track	select; required; empty value for Choose…	Web	Choose…	Passed
Arrival day	date; 2026-11-20…2026-11-22; required	21 Nov 2026	25 Nov 2026	Passed
Check-in time	time; 09:00-18:00; step=1800; required	10:30	10:15; 19:00	Passed
Entry fee	number; 0-5000; step=500; required	1500	1250; 6000	Passed
Experience	radio; shared name; required	Beginner	nothing selected	Passed
GitHub profile	url; optional	empty; https://github.com/yerassyl-dev	github.com/yerassyl-dev	Passed
Password	password; 8-20 characters; required	Astana2026	short	Passed
Motivation	textarea; 50-300 characters; required	120 characters	49 characters	Passed
Rules agreement	checkbox; required	checked	unchecked	Passed

All values from the assignment produced the expected results. Boundary values and empty required fields were also tested. All 64 value tests passed. For time inputs, step=1800 means 30 minutes; maxlength prevents typing extra characters.

Task 3. Improved cafe reservation form

I kept all fields from the original form and added labels, unique id values, name attributes, and appropriate input types. The Inside and Terrace radio buttons share a name. A real submit button replaces the div.



Verification

The form works with the keyboard: Tab moves focus, arrow keys select a radio option, Space toggles the agreement, and Enter submits the form. There is no horizontal scrolling at 375 or 1280 px.

The official W3C Validator found no errors or warnings in task1.html, task2.html, or task3.html. The local demo handler only confirms submission; it does not create a real reservation.

Task 3. 10 problems in the original form

Line numbers refer to the original snippet from the assignment PDF, saved in report/original-task3.txt.

No. / lines	Original code or issue	Why it is a problem	Fix
1 3-4	<input type="text" placeholder="Your name">; <input type="text" placeholder="Email">	Name and email placeholders disappear during input.	Added labels with matching for/id values.
2 3-10, 14, 18-19	Text inputs, select, textarea and checkbox lack name attributes.	Values without a name are omitted from form submission.	Added a name to every field.
3 4	<input type="text" placeholder="Email">	type=text accepts invalid email addresses.	Changed to type=email.
4 5-6	<label>Phone</label>; <input type="text" id="phone">	Clicking Phone does not focus its input.	Added for=guest-phone and a matching id.
5 7-8	<label for="guests">Guests</label>; <input type="text" id="people">	Clicking Guests does not focus its input.	Both the label's for and input's id use guests.
6 8	<input type="text" id="people">	The guest count accepts letters and fractions.	Set type=number, min=1 and step=1.
7 10	<input type="text" id="date" placeholder="dd/mm/yyyy">	A text input does not validate dates.	Changed to type=date.
8 10, 12	id="date" is shared by the date input and Inside radio.	A duplicate id makes label associations ambiguous.	Assigned unique ids to the date input and radio.
9 12-13	The radios use name="inside" and name="terrace".	Inside and Terrace can be selected together.	Shared name=seating, distinct values and a fieldset.
10 20	Reserve	A div neither submits the form nor acts as a keyboard button.	Replaced with button type=submit.

Alternatives and conclusion

Lines 3-4

Alternative. Wrap each input in its label to create an implicit association without for.

Why I chose this fix. Explicit for/id links let labels and inputs occupy separate Bootstrap grid cells while preserving accessible names and click-to-focus behavior.

Lines 12-13

Alternative. Replace the two radios with a select offering Inside and Terrace.

Why I chose this fix. Radios keep both choices visible, preserve the original control type and support selection with arrow keys.

Conclusion

In this assignment, I learned how to build forms with the Bootstrap grid, validate data using HTML attributes, and connect fields to labels. The validation rules are defined in HTML, while CSS controls the appearance and error feedback.

Checks and setup

All 45 functional checks and 64 value tests passed. The pages contain no JavaScript, pattern, or novalidate. Open index.html to view the forms. To demonstrate successful submission, run python3 scripts/serve.py and open http://127.0.0.1:8000/.

GitHub

https://github.com/mmionya/web-technologies-assignment4

GitHub Pages

https://mmionya.github.io/web-technologies-assignment4/

Sources

Bootstrap: Forms

MDN: :user-invalid

MDN: input time

W3C Validator

The requirements, train form reference, and original cafe form come from Assignment 4.pdf.